Chemistry - Undergraduate Level Assessment 19 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Of the following ions, which has the smallest radius?
   (a) K+
   (b) Ca_2+
   (c) Sc_3+
   (d) Rb+
2. Which of the following is the most common naturally-occurring form in which silicon is found?
   (a) Metallic element
   (b) Sulfide
   (c) Fluoride
   (d) Oxide
3. A set of hybrid sp^3 orbitals for a carbon atom is given above. Which of the following is NOT true about the orbitals?
   (a) The orbitals are degenerate.
   (b) The set of orbitals has a tetrahedral geometry.
   (c) These orbitals are constructed from a linear combination of atomic orbitals.
   (d) Each hybrid orbital may hold four electrons.
4. Elements with partially filled 4 f or 5 f orbitals include all of the following EXCEPT
   (a) Cu
   (b) Gd
   (c) Eu
   (d) Am
5. Which of the following experimental observations were explained by Planck's quantum theory?
   (a) Blackbody radiation curves
   (b) Emission spectra of diatomic molecules
   (c) Electron diffraction patterns
   (d) Temperature dependence of reaction rates

True / False (10 marks)

Answer True or False

6. True or False: A magnetic field of 0.5 T can induce the same level of polarization for 13 C at 298 K.

7. True or False: The molecular geometry of thionyl chloride, SOCl 2, is best described as trigonal pyramidal.

8. True or False: Most organic functional groups absorb in a characteristic region of the infrared spectrum, aiding in structural analysis.

9. True or False: The strongest base in liquid ammonia is the amide ion, NH 2-.

10. True or False: Hexane has five distinct chemical shifts in its 13 C spectrum due to its symmetrical structure.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process by which cobalt-60 is produced from cobalt-59 through neutron bombardment, and analyze its significance in radiation therapy for cancer treatment.

12. Discuss the EPR spectrum of a rigid nitronyl nitroxide diradical with J << a, focusing on how the coupling of unpaired electrons to nitrogen nuclei affects the number of spectral lines produced.

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